

CUMULATIVE RADIATION EFFECT PART I: FRACTIONATED TREATMENT REGIMES

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The object of this paper is to establish a criterion as a basis on which to assess the biological effect of the widely varied fractionated regimes at present used in the radiotherapy of malignant diseases. These assessments are based on a scale of the accumulated sub-tolerance radiation damage, the Cumulative Radiation Effect (CRE). The CRE is a generalised form of the empirical function (NSD) for the assessment of damage at tolerance, which has been described by Ellis. The properties of the CRE function are examined in some detail, and its application to various clinical situations described and illustrated by numerical examples. The CRE can be considered as a summation of the effects of fractions or groups of fractions, but these effects cannot be evaluated without consideration of the amount of damage caused by previous fractions. To take previous damage into account, the equivalence conditions existing at the junctions between fractions or schedules must be determined. It is found that if a particular regime achieves a certain CRE, its individual fractions may be given in any order whatsoever, without altering the CRE attained.

1. INTRODUCTION

ONE of the many pressing problems confronting the clinician in the practice of radiotherapy is the rationalisation of radiation fractionation regimes in the treatment of malignant diseases. These regimes vary considerably from one centre to another, and it is most essential to find some unifying criterion as a basis on which to assess and compare them. Ideally, such a criterion would provide a measure of radiation damage to both normal and tumour tissue for a particular regime of treatment, allowing not only the comparison of the biological effects of different regimes, but also the selection of the most beneficial form of treatment for a particular lesion.

A significant contribution to a rationalisation of treatment regimes has been made by Ellis and his colleagues (Ellis, 1967; Ellis, 1968; Ellis, 1969; Winston, Ellis and Hall, 1969), who have shown that one figure, representing the normal connective tissue tolerance, can be used to designate any course of radiation which reaches that tolerance, the practical upper limit of any radiation schedule. Ellis has termed this figure the NSD (nominal standard dose), and has shown that it can be expressed by

$$\text{NSD} = (\text{Total Dose}) \cdot N^{-0.24} \cdot T^{-0.11}, \quad (1)$$

where the total dose is in rads, N is the number of

fractions and T is the total treatment time. This equation is based on the iso-effect curves for different fractionation schedules for squamous cell carcinoma, skin erythema and normal tissue tolerance, as shown in Figure 1 and taken from Cohen (1960), but, however, applies only to normal tissue at the limit of tolerance.

This approach of Ellis to the rationalisation of fractionated treatment regimes signifies a real advance, but is to some extent a restricted interpretation of the available data. The object of this paper, and of succeeding papers, is to extend the work of Ellis by deriving as general a formulation as possible from the available clinical data, and by taking the formulation to some logical conclusions whose verification will serve as a test for the completeness of the description of clinical situations given by it.

It must be stressed that this type of approach does not fulfil all the requirements of the criterion mentioned above, as it takes no quantitative account of the difference in the effects of radiation on normal and tumour tissues. Ellis' statement that 'the general aim must be to achieve the normal tissue tolerance in as short a time as possible' (Ellis, 1969) is certainly an oversimplification and, indeed, Cohen's graphs indicate that the differential effect becomes more favourable the longer the treatment lasts. The removal of this limitation of the approach

and also that imposed by its empirical nature awaits a greater knowledge and understanding of the effects of radiation on normal and tumour tissue *in vivo*.

2. THE CUMULATIVE RADIATION EFFECT

In order to extend and generalise the approach of Ellis to the rationalisation of radiotherapy treatment regimes, it is essential to appreciate that equation (1), with the derivation of which the present authors are in agreement, defines a particular value of a more general and biologically significant function, which gives important information on sub-tolerance damage to normal connective tissue. In his treatment of the function, Ellis assigned biological significance to it only at its upper limit (NSD), which is defined by tissue tolerance. It is our premise that Ellis' equation can be applied to any radiation effect on normal tissue up to, and including, tolerance.

The equation describing the iso-effect curves for fractionated radiation (equation (1)) can be written as a proportionality relation:

$$D \propto N^{0.24} \cdot T^{0.11}.$$

In an alternative form, this equation can be written as

$$D = R_F \cdot N^{0.24} \cdot T^{0.11}, \quad (2)$$

where R_F is a positive constant of proportionality. We suggest that as R_F assumes different values, this equation represents the iso-effect curves of a variety of radiation effects, the value of R_F increasing with the severity of the effect.

This additional hypothesis is supported by the fact that the gradients of the iso-effect curves for erythema and normal connective tissue tolerance are the same within the limits of clinical discrimination, although, of course, their intercepts, R_F , are different (Fig. 1). Since, in addition, erythema is a lesser radiation effect than tolerance and its value of R_F is smaller than that for tolerance, it is a reasonable supposition, which could be verified by subsequent experiments, that the iso-effect curves of other radiation effects intermediate between none and normal connective tissue tolerance have the same gradient with their R_F values increasing monotonically with the degree of damage.

Since R_F is essentially a measure of the accumulated biological effect of successive doses of radiation, we propose that it be termed the 'Cumulative Radiation Effect'. The nominal standard dose of Ellis then becomes the value of the Cumulative Radiation Effect at tolerance, a special value of the more general function.

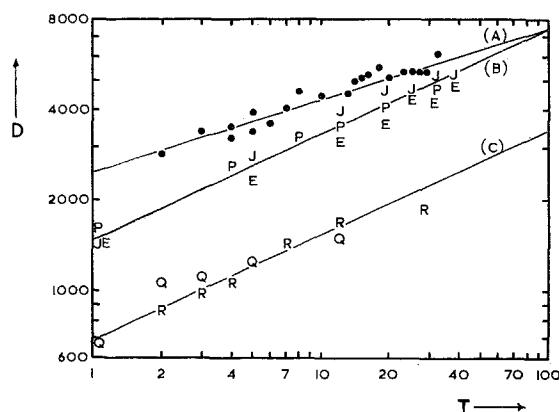


FIG. 1

Iso-effect curves for different fractionation schedules shown as the dependence of the total dose (D) in rads on the total treatment time (T) in days for the following effects: squamous cell carcinoma (A), normal connective tissue tolerance (B) and skin erythema (C). [Taken from Cohen (1960) and Ellis (1969).]

The choice of R_F as a measure of radiation effect is not the only possible one, since any positive power of R_F would be equally valid. Any categorical statement as to which power is 'correct' is impossible with our present lack of understanding of the nature of radiation damage. From equation (2) it can be seen that, if N and T are kept constant, R_F is proportional to the total dose, D . Our choice of R_F itself as a measure of damage is governed by two considerations:

- (i) it seems reasonable to use R_F to unit power, i.e. to define the effect to be proportional to the total dose;
- (ii) this choice of effect function facilitates the extension of the method to continuous radiation (to be described in a later paper).

As will be shown later, Ellis' partial tolerance measure of damage is effectively a positive power of R_F , multiplied by a normalising constant.

It can be seen from the above equation (2) for the CRE (Cumulative Radiation Effect) that it has the unit of $1 \text{ rad} \cdot \text{day}^{-0.11}$ or, even more explicitly, $100 \text{ ergs} \cdot \text{gm}^{-1} \cdot \text{day}^{-0.11}$. This is a most cumbersome unit for frequent use and so, for convenience, it is proposed that it should be replaced in practice by 'reu' which is suggested by the initial letters of 'radiation effect unit'. It has been decided not to call the unit of the CRE the 'ret' (rad equivalent therapy), as suggested by Ellis, because the unit of effect, as specified above, has different dimensions from the rad and is not analogous to the 'rem'. The term 'radiation effect unit' is felt to be more self-explanatory and less confusing than 'rad equivalent therapy', although no fundamental significance is claimed for the 'reu'—it is merely a convenient shorthand for our present purposes.

The remainder of this paper will be devoted to the development of our more general concept of the CRE, involving a discussion of the properties of the function, R_F , and some of its applications to fractionated treatment regimes. Throughout this discussion, it will be assumed that Co^{60} γ -radiation or the equivalent X-radiation is being considered, although the argument can be applied to other qualities of radiation if the dose is adjusted to take account of the quality factor. This consideration and others such as the dependence of the CRE on the size of the treated volume or area will be discussed in a later paper.

3. GENERAL INFERENCES FROM THE CRE EQUATION

In this section and in those that follow, the terms 'schedule' and 'regime' will be used in the following sense.

A *schedule* of treatment is the repetitive application of fractions of equal doses at equal time intervals over a specified treatment time.

A *regime* of treatment is a systematic course of radiotherapy comprising one or more schedules.

The expression for the CRE can be written in the form:

$$R_F = D \cdot N^{-0.24} \cdot T^{-0.11}. \quad (3)$$

This equation can be rewritten in a more convenient form for analysis by defining

$d = D/N$ as the dose per fraction, and

$x = T/N$ as the ratio of total time (inclusive of the first and last days of treatment) to the number of fractions, so that

$$\begin{aligned} R_F &= d \cdot N \cdot N^{-0.24} \cdot x^{-0.11} \cdot N^{-0.11} \\ &= x^{-0.11} \cdot d \cdot N^{0.65}. \end{aligned}$$

As a further simplification, convenient for later discussion, let

$$R_F = \rho \cdot d \cdot N^m, \quad (4)$$

where $\rho = x^{-0.11}$ and $m = 0.65$. ρ then describes the time interval of the repetition of the fractions of the treatment schedule under consideration, and will be referred to as the 'repetition parameter'.

The relationship of our approach to that of Ellis can be demonstrated from a consideration of equation (4). Whereas, on our definition of CRE, R_F is proportional to the total dose, D , Ellis' choice of a scale of damage was determined by the criterion that effect should be proportional to N , with ρ and d constant. From equation (4) it can be seen that $R_F^{1/m}$ fulfils this stipulation. However, at tolerance,

this function would have a different value from that given by R_F from equation (4), i.e. NSD, as previously defined by Ellis, and so $R_F^{1/m}$ must be multiplied by a normalisation constant. This constant can be shown to be $R_{FT}^{(1-1/m)}$ where R_{FT} is the tolerance value of R_F . Thus, Ellis and his co-workers have defined a scale of damage which differs from ours at all levels of damage except tolerance. Since the normalisation constant depends on the tolerance value of the CRE, any theoretical simplicity of calculation Ellis may have gained by this definition of scale is in practice offset by the necessity to refer, at every stage of a calculation, to the tolerance condition for the schedule under consideration.

In order to examine the behaviour of equation (4), it is instructive to observe the way in which the CRE responds to changes in one of the three variables ρ , d or N , while the others are held constant. If ρ and d are kept constant and only N allowed to vary, equation (4) not only describes the variation of the final CRE with the total number of treatments for different schedules, but also shows the way in which, for any one schedule, the value of R_F increases with the number of fractions administered, from the inception of treatment until the completion of the course. The CRE achieved for increasing numbers of fractions, N , is shown in Fig. 2, where it can be seen that the greatest contribution to the CRE is made by the earliest fractions. It should be stressed that the CRE does not necessarily have a linear relationship to any other measure of radiation damage such as the fraction of cells killed.

The variation in the CRE with the number of fractions can be considered from another point of view. Consider the case in which the dose per fraction, d , and the repetition parameter, ρ , are

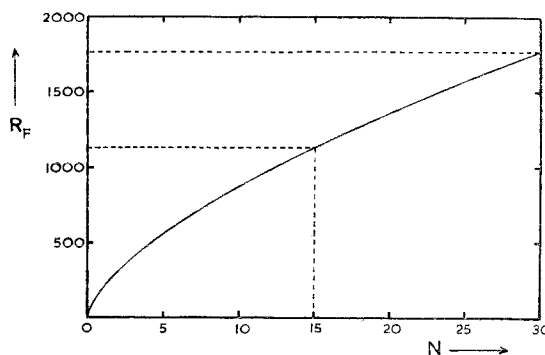


FIG. 2

Relationship between the CRE (R_F) in reu and the number of fractions (N) given of a schedule, when the repetition parameter, ρ , and the dose per fraction, d , are fixed (in this case, $\rho = 0.97$ and $d = 200$ rads).

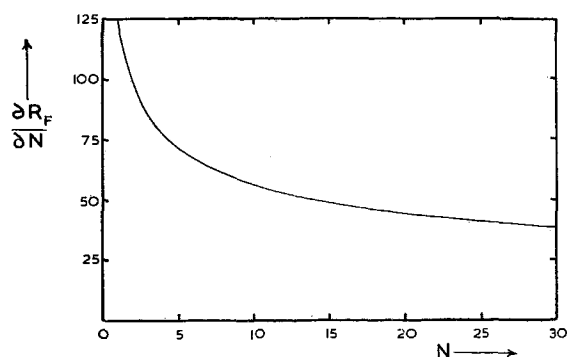


FIG. 3

Relationship between the rate of change of CRE ($\partial R_F/\partial N$) and the number of fractions (N) given of a schedule, when the repetition parameter, ρ , and the dose per fraction, d , are fixed (in this case, $\rho = 0.97$ and $d = 200$ rads).

held constant. It is desired to examine the manner in which the CRE is built up by successive radiation fractions. The CRE attained by a number of fractions, N , is given by

$$R_F = \rho \cdot d \cdot N^m.$$

In practice, each of the N sequential fractions contributing to the ultimate CRE makes a different contribution. The rate of change of CRE with N can be written as

$$\frac{\partial R_F}{\partial N} = m \cdot \rho \cdot d \cdot N^{m-1}, \quad (5)$$

and is shown graphically in Fig. 3. The rapid drop in the value of $\frac{\partial R_F}{\partial N}$ clearly indicates that later

fractions of a treatment schedule contribute progressively less to the ultimate value of the CRE. The contribution to the CRE in a fractional interval, dN , is

$$\frac{\partial R_F}{\partial N} \cdot dN = m \cdot \rho \cdot d \cdot N^{m-1} \cdot dN.$$

Therefore, the contribution to the CRE by the fifth fraction, for example, is

$$\begin{aligned} \int_4^5 m \cdot \rho \cdot d \cdot N^{m-1} \cdot dN &= [\rho \cdot d \cdot N^m]_4^5 \\ &= \rho \cdot d \cdot (5^m - 4^m). \end{aligned}$$

This result can, in fact, be written down directly from equation (4), but the above derivation is instructive.

If this analysis is continued, it is found that the CRE can be represented as a summation of effects following on each other, and can be written as

$$\begin{aligned} R_F &= \rho \cdot d \cdot [(-0^m + 1^m) + (-1^m + 2^m) + \dots \\ &\quad + (-\{N-1\}^m + N^m)] \\ &= \rho \cdot d \cdot N^m. \end{aligned}$$

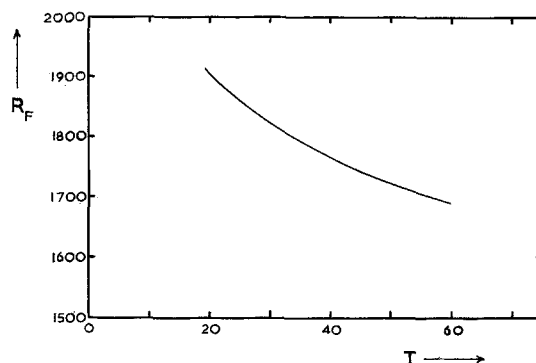


FIG. 4

Relationship between the CRE (R_F) in reu and the total treatment time in days when the number of fractions, N , and the dose per fraction, d , are fixed (in this case, $N = 30$ and $d = 200$ rads).

The CRE could equally well be subdivided into groups of fractions so that it could be written as

$$R_F = \rho \cdot d \cdot [(-0^m + 2^m) + (-2^m + N^m)] = \rho \cdot d \cdot N^m.$$

The effect due to a fraction or a group of fractions cannot be considered alone, but must be regarded in the light of the history of any previous radiation. The CRE equation includes the effects of cellular recovery after irradiation both by intracellular repair and homeostatic control. It would thus be expected that the effect of a fraction should depend upon the accumulated effect of previous fractions.

An implicit assumption in the discussion until now has been that the courses of radiation considered comprise only one schedule, i.e. for the whole course, ρ and d are constant. The formulation can be extended to include regimes of more than one schedule, each having different values of ρ and d , by a technique similar to the above—equation (5) is integrated over each group of fractions from the same schedule, and the resultant increments summed. This process involves a consideration of the conditions existing at the junction points between adjacent groups, and is fully discussed in Section 4.

The effect on the CRE of altering the repetition parameter of a treatment schedule is demonstrated in Fig. 4 as the dependence of the CRE on changes in the total treatment time. When the number of treatments is held constant, ρ is directly related to the treatment time since

$$\rho = x^{-0.11} = (T/N)^{-0.11}.$$

It can be seen that when the same number of treatments is compressed into a shorter time, the schedule is more effective. This is a familiar feature in clinical radiotherapy.

The fact that the CRE is proportional to the total dose, D , when the number of fractions and the total time are fixed (equation (3)) implies that, under these conditions, the CRE is no more fundamental than the total dose in estimating the relative effect of treatment schedules. The great value of the CRE lies in its ability to compare various regimes of treatment by providing a scale of radiation effect, on which an estimate of therapeutic benefit can be based.

4. APPLICATIONS OF THE CUMULATIVE RADIATION EFFECT

For practical applications, it is clearly essential to appreciate the ways in which the CRE equation can be used to find the CRE achieved by various treatment regimes. The basic principles required for such applications will be developed and elaborated in this and the following sections.

In radiotherapy, courses of treatment are often given by alternate fractions or groups of fractions of different schedules. Such a regime may have been intended from the inception of treatment or may have been forced upon the therapist by deterioration in the patient's condition. Such possibilities are demonstrated in the following example, to which further reference will be made in subsequent sections.

A patient having carcinoma of the ovaries is to be treated by two parallel opposed fields given alternately on a daily basis. Owing to recent surgery, the patient cannot lie prone and, for the first week of a 4-week therapy regime, must be treated by an anterior field alone. In the second week, the posterior field is given alone to balance the treatment doses, and, in the second fortnight, the originally intended alternate fractions are resumed. In estimating the CRE achieved with more than one treatment schedule, such as in this example, it is most important to consider the conditions existing at the junction between schedules and fractions.

(a) TWO SEQUENTIAL SCHEDULES: JUNCTIONS BETWEEN SCHEDULES AND FRACTIONS.—Take the case in which two entirely different schedules, A and B, are to be given sequentially to achieve a desired CRE, R_F . Let the two repetition parameters and doses per fraction be ρ_a , d_a and ρ_b , d_b , respectively. If each schedule were given alone for the total time, the numbers of fractions of each schedule required to achieve the same CRE, R_F , would be N_a and N_b , respectively, such that

$$R_F = \rho_a \cdot d_a \cdot N_a^m = \rho_b \cdot d_b \cdot N_b^m.$$

Similarly, some intermediate CRE, r , could be achieved by the numbers of fractions, n_a and n_b , respectively, of the same schedules, so that

$$r = \rho_a \cdot d_a \cdot n_a^m = \rho_b \cdot d_b \cdot n_b^m.$$

From the earlier discussion, it follows that, for two sequential groups of fractions of the same schedule, the effect of the second group cannot be evaluated without a knowledge of the effect of the first group, which provides a starting-point or initial condition for the subsequent accumulation of damage. Thus, the increment of CRE required to make up the difference between the CREs, r and R_F , can be written as

$$R_F - r = \rho_a \cdot d_a \cdot (N_a^m - n_a^m) = \rho_b \cdot d_b \cdot (N_b^m - n_b^m).$$

If, on the other hand, schedule A were to be maintained to a CRE, r , and then the treatment continued, without any gap, on schedule B to achieve the CRE, R_F , it is again the case that the effect of the fractions of schedule B must be added to, and take account of, the CRE, r , obtained with schedule A. From the above equations, it can be seen that

$$R_F = \rho_a \cdot d_a \cdot n_a^m + \rho_b \cdot d_b \cdot (N_b^m - n_b^m),$$

where the two terms on the right-hand side represent the effects of the first and second schedules, respectively. There is no necessity on mathematical grounds for the number of fractions to be integral. Thus, the number of treatments required on each schedule used sequentially to achieve the desired CRE, R_F , can be found and is n_a on the first schedule and $(N_b - n_b)$ on the second.

The important point is to determine the equivalence condition at the junction between schedules or fractions of a regime, which in this case is

$$\rho_a \cdot d_a \cdot n_a^m = \rho_b \cdot d_b \cdot n_b^m.$$

From the above equations, it can also be seen that

$$(N_a/n_a)^m = (N_b/n_b)^m,$$

so that

$$N_a/n_a = N_b/n_b.$$

The application of this discussion to practical cases can, perhaps, be more easily appreciated in the following example.

Example 1.—It was intended to treat a patient using Co^{60} γ -radiation according to a schedule of treatment giving 5,000 rads in 4 weeks. As a result of patient reaction, it was decided at the end of 3 weeks to reduce the rate of radiation by giving 3 fractions per week instead of 5 and by reducing the dose per fraction from 250 to 200 rads for the

remainder of the treatment. The problem, then, is to determine the number of fractions required on the new schedule to achieve the originally desired CRE.

For ease of reference, let the first and second schedules be termed schedules A and B, respectively. The repetition parameters and the doses per fraction of these two schedules are $\rho_a = 0.97$, $d_a = 250$ rads, and $\rho_b = 0.92$, $d_b = 200$ rads, respectively. The values of ρ_a and ρ_b depend to some extent on the length of treatment and the day of the week on which treatment commences. This aspect of the problem has been discussed at some length by Winston *et al.* (1969). These variations in the values of ρ_a and ρ_b are in general slight for the schedules under consideration and have been ignored for simplicity in this and subsequent examples, the ρ 's being considered as constants for any one schedule.

By direct application of the CRE formula (equation (4)), it can be seen that the CRE, R_F , achieved by schedule A had it been given for the full treatment, as originally intended, to give 5,000 rads in 20 fractions in 4 weeks (26 days) would have been 1,700 reu. On the other hand, the same CRE, R_F , would have been attained by schedule B given for the full treatment by giving 6,140 rads in 30.7 fractions in just over 10 weeks (71 days). The intermediate CRE, r , achieved by schedule A at the end of the first 3 weeks when this schedule had to be abandoned due to patient reaction is readily seen, by application of the CRE formula, to be 1,410 reu, given in 15 fractions.

To find the number of fractions to be given on schedule B to complete treatment, it is necessary to establish the equivalence condition existing at the junction between schedules A and B at the CRE, r . It was shown above that this condition is

$$\begin{aligned}\rho_a \cdot d_a \cdot n_a^m &= \rho_b \cdot d_b \cdot n_b^m, \\ \text{or } 0.97 \cdot 250 \cdot 15^{0.65} &= 0.92 \cdot 200 \cdot n_b^{0.65}, \\ \text{so that } n_b &= 23.0.\end{aligned}$$

Schedule B must then be maintained for

$$N_b - n_b = 30.7 - 23.0 = 7.7 \text{ fractions,}$$

which will be given in just under 3 weeks (17 days) at 3 fractions per week. It can be readily seen that the subsidiary condition

$$N_a/n_a = N_b/n_b = 1.33$$

is also satisfied.

It is a natural sequel to the analysis described above on the equivalence conditions existing at the

junctions between schedules and fractions to consider the CRE achieved when two schedules A and B, which are specified as follows, are given sequentially without any gap in treatment: these schedules have the same repetition parameter, ρ , and the same number of fractions $N_a = N_b = \mathcal{N}/2$, where $\mathcal{N}$ is the total number of fractions of both schedules, i.e. the total number of the regime, but different doses per fraction, d_a and d_b , respectively. This situation is frequently encountered clinically and is illustrated by the treatment given (to tissue not in the mid-plane) in the first 2 weeks to the patient whose case was described at the beginning of section 4 above.

By use of the equivalence condition at the junction between the schedules, the CRE achieved by this regime of treatment can be shown to be

$$R_F = \rho \mathcal{N}^m \left[\frac{d_a^{1/m} + d_b^{1/m}}{2} \right]^m.$$

A proof of this result is given in Appendix 1. It is of interest to note that the equation for the CRE achieved by the regime of sequential schedules is of the same form as the simple or basic CRE equation, in which the dose per fraction, d , has been replaced by the expression for the 'effective dose per fraction'

$$\left[\frac{d_a^{1/m} + d_b^{1/m}}{2} \right]^m.$$

From the symmetry of this expression and, of course, of that for the CRE, R_F , attained by this regime of sequential schedules, it is clear that the same CRE is achieved regardless of which schedule is given first.

(b) REGIMES OF TWO SCHEDULES WHOSE FRACTIONS ARE GIVEN ALTERNATELY.—The foregoing analysis of two different schedules of treatment given sequentially can obviously be extended by repeated application, in the limiting case of schedules of one fraction, to the case of alternating fractions from two schedules. As a specific case of this, which is the one most frequently encountered in practice, consider a treatment of the same number of alternating fractions from two schedules A and B having the same repetition parameter and the doses per fraction d_a and d_b respectively. The number of fractions in each schedule is $N_a = N_b = \mathcal{N}/2$, where $\mathcal{N}$ is the total number of fractions of both schedules, i.e. the total number of the regime. This situation is also illustrated by the treatment given (to tissue not in the mid-plane) in the second fortnight to the patient with carcinoma of the ovaries

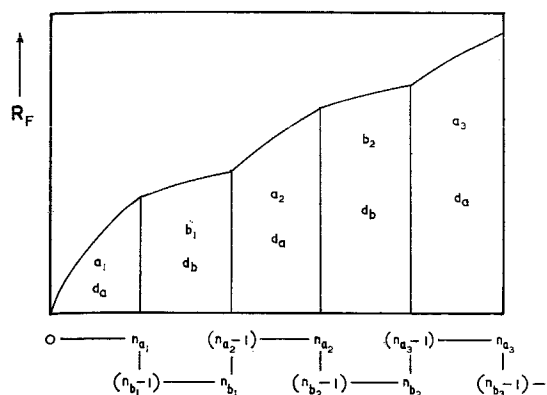


FIG. 5

Accumulation of the CRE (R_F) with successive fractions given alternately from two schedules A and B (symbols explained in Appendix 2).

whose case was described at the beginning of Section 4.

The manner in which the CRE could be accumulated is shown in Fig. 5, where the first fractions to be given of schedules A and B are labelled a_1 and b_1 , respectively, and so on, other fractions being similarly identified. The effect of each fraction given must be added to the accumulated radiation damage of previous fractions, and so radiation effects can be related by the equivalence conditions at the junctions between fractions. By the application of the properties of the Cumulative Radiation Effect outlined above, it can be shown (see Appendix 2) that the CRE, R_F , which can be achieved by a regime of alternate fractions of schedules A and B is

$$R_F = \rho_e \mathcal{N}^m \left[\frac{d_a^{1/m} + d_b^{1/m}}{2} \right]^m \quad (6)$$

where it is convenient to define ρ_e as an 'effective repetition parameter', which would be the repetition parameter if the treatment consisted of one schedule lasting for the same time and containing the same number of fractions as both schedules together. For this particular case of a regime of an equal number of alternating fractions, the effective repetition parameter, ρ_e , has the same numerical value as the repetition parameter, ρ , in the previous section for sequential schedules when the time interval between adjacent fractions is the same. From the symmetry of equation (6), it is clear that the same CRE is achieved regardless of which dose is given first.

It can also be seen that this result is the same as that found for sequential schedules. It is most interesting, and perhaps surprising, that the CRE for two sequential schedules is the same as that for

a regime consisting of the fractions of the original schedules given alternately.

The application of this analysis to regimes of two schedules whose fractions are given alternately can be demonstrated by the following example, in which the numerical values are taken from a letter on the two-field treatment of breast carcinoma by Ellis, Winston, Fowler and DeGinder (1969).

Example 2.—It was intended to treat radically a carcinoma of the breast after simple mastectomy by tangentially opposed fields on a schedule of 3 fractions per week over a treatment time of 5 weeks. Each field was to be given on alternate days, so that the central portion of the treated volume received 4,400 rads in 16 fractions, each of 275 rads, while the region near the edge of the volume received 4,000 rads in 8 fractions (d_a equals 500 rads) and 1,360 rads in 8 fractions (d_b equals 170 rads).

It was found in Example 1 that, for a schedule of 3 fractions per week, $\rho = 0.92$ and so, from the above analysis, the effective repetition parameter, $\rho_e = 0.92$. The CRE achieved at the centre of the volume can be found from equation (4) and is

$$R_F = 0.92 \cdot 275 \cdot 16^{0.65} = 1,535 \text{ reu.}$$

The CRE achieved near the edge of the volume can be found using equation (6) and is

$$\begin{aligned} R_F &= \rho_e \mathcal{N}^m \left[\frac{d_a^{1/m} + d_b^{1/m}}{2} \right]^m \\ &= 0.92 \cdot 16^{0.65} \cdot \left[\frac{500^{1.54} + 170^{1.54}}{2} \right]^{0.65} \\ &= 2,000 \text{ reu.} \end{aligned}$$

It can be seen that the CRE near the edge of the treated volume is considerably greater than that at the centre, as would be expected.

(c) GENERAL CASE FOR THE EQUIVALENCE OF PERMUTED FRACTIONS.—The conclusion that the CRE achieved by two schedules A and B is independent of the order of the schedules and does not change if the fractions of the original schedules are given alternately raises the suspicion that some general equivalence of permuted fractions or groups of fractions may exist. In consequence, it is of interest to calculate the increases in the CRE from some value, which has been achieved by some arbitrary schedule of treatments, by the application of 2 different fractions or groups of fractions in reverse orders. The repetition parameter, ρ , is again considered constant. By the application, once again, of the equivalence condition at the

junction between fractions, it is easily shown that the order of application of 2 fractions or groups of fractions given at some arbitrary point in a course of radiation treatment has no effect on the CRE achieved. The proof of this argument is outlined in Appendix 3. By repeated application of this result, the equivalence of any two permutations, and thus all permutations, of a given group of fractions can readily be shown.

A firm conclusion from the preceding arguments and discussion is that if a particular regime achieves a desired CRE, its individual fractions can be given in any order whatsoever without altering the ultimate CRE attained, provided the interval between successive fractions remains constant. Although the previous sections have accentuated the case of two schedules as the simplest and that most frequently encountered in practice, the general proof for the equivalence of permuted fractions justifies the application of the principles outlined to more than two schedules.

The situation examined in Example 1 provides a simple numerical verification of the effect produced by a reversal in the order of the application of groups of fractions. It was shown in that example that a CRE of 1,700 reu could be achieved by giving 15 fractions on schedule A followed by 7.7 fractions on schedule B. To demonstrate the validity of reversing the order of application, it is necessary to show that the same CRE of 1,700 reu is attained when the 7.7 fractions on schedule B are given first and the 15 fractions of schedule A second.

Since schedule B is given first, a different intermediate CRE, s , is found, such that

$$s = \rho_b \cdot d_b \cdot n_b'^m = 0.92 \cdot 200 \cdot 7.7^{0.65} \\ = 694 \text{ reu.}$$

To determine the effect of the fractions of schedule A, the radiation damage already generated by schedule B must be taken into account and so the equivalence condition at the intermediate CRE, s , must be established, and is

$$\rho_b \cdot d_b \cdot n_b'^m = \rho_a \cdot d_a \cdot n_a'^m.$$

The number of fractions of schedule A which would have generated the same damage as the 7.7 fractions of schedule B is

$$n_a' = \left[\frac{\rho_b \cdot d_b \cdot n_b'^m}{\rho_a \cdot d_a} \right]^{1/m} \\ = \left[\frac{694}{0.97 \times 250} \right]^{1.54} = 5.0$$

Therefore, as 15 fractions are to be given on schedule A after this intermediate CRE, s , has been reached, the total number of fractions which would have effectively been given on schedule A would be 20.0. By application of the CRE formula (equation (4)), the ultimate CRE achieved is seen to be

$$R_F = 1,700 \text{ reu,}$$

which is the same as that obtained with the original order of schedules. From this example, it is thus clear that the reversal in the order of the application of the groups of fractions from the two schedules A and B is immaterial to the ultimate CRE achieved.

5. CONCLUSIONS

From a consideration of clinical and experimental iso-effect curves, it has been found possible to define a function, the Cumulative Radiation Effect (CRE), which gives a measure of accumulated biological effect on normal connective tissue at all sub-tolerance levels of damage. An important consequence of this approach is that the effect of any fraction or group of fractions cannot be evaluated without consideration of the amount of damage caused by previous fractions. This is equivalent to saying that, in order to evaluate the increment in the CRE caused by a part of a regime, it is essential to have a knowledge of the initial conditions, i.e. those existing before this additional part, or schedule, is given. These conditions can be established by the specification of the junction point between the two parts of the regime in terms of the number of fractions of the second schedule which it would have been necessary to give in order to reach the CRE attained by the first part of the regime.

One interesting result of the formulation is that, for a given set of equally spaced fractions, the order in which the fractions are administered has no effect on the final CRE achieved.

This work has been limited to the study of fractionated treatment regimes, but the extension of this approach to continuous radiation therapy and more complex modes of treatment will be discussed in following papers. It is hoped that this generalised approach to the rationalisation of radiation therapy will make a real contribution to the solution of clinical problems and the improvement of patient therapy.

Appendix 1: Two Different Sequential Schedules

Consider the case in which two treatment schedules A and B are given sequentially to achieve a desired CRE, R_F . It is assumed that both

schedules have the same repetition parameter, ρ , and the same number of fractions, $N_a = N_b$, but different doses per fraction, d_a and d_b , respectively.

At the junction between schedules A and B, the following equivalence condition must hold at the intermediate CRE:

$$\rho \cdot d_a \cdot n_a^m = \rho \cdot d_b \cdot n_b^m,$$

where $n_a = N_a$, and n_b is the number of fractions of schedule B which would be required to achieve the same intermediate CRE as n_a fractions of schedule A. Since the CRE can be represented as a summation of the effect of schedule B following on schedule A, the ultimate CRE, R_F , can be written as:

$$R_F = \rho \{d_a \cdot N_a^m + d_b \cdot [(n_b + N_b)^m - n_b^m]\}$$

Using the equivalence condition and the fact that $N_b = N_a = n_a$, and cancelling the first and last terms,

$$\begin{aligned} R_F &= \rho d_b \cdot [n_a(d_a/d_b)^{1/m} + N_a]^m \\ &= \rho [N_a \cdot d_a^{1/m} + N_a \cdot d_b^{1/m}]^m \\ &= \rho N_a^m [d_a^{1/m} + d_b^{1/m}]^m. \end{aligned}$$

Let $\mathcal{N}$ be the total number of fractions given in the treatment; then,

$$\begin{aligned} \mathcal{N} &= 2N_a, \\ \text{and } R_F &= \rho \mathcal{N}^m \left[\frac{d_a^{1/m} + d_b^{1/m}}{2} \right]^m. \end{aligned}$$

Appendix 2: Regimes of Two Schedules whose Fractions are given Alternately

Consider the situation in which the fractions of two different treatment schedules A and B are given alternately to achieve a desired CRE, R_F , the combined schedules having the 'effective repetition parameter', ρ_e , which would be the repetition parameter if the treatment consisted of one schedule lasting for the same time and containing the same number of fractions as both schedules together. These two schedules are assumed to have the same number of fractions, $N_a = N_b = N$, where the subscripts a and b refer to the schedules A and B, respectively, and are added for clarity. Let the doses per fraction for the schedules A and B be d_a and d_b , respectively. Succeeding fractions of schedules A and B are labelled $a_1, a_2, a_3, \dots, a_N$, and $b_1, b_2, b_3, \dots, b_N$, respectively, and the numbers of fractions, which would, with the repetition parameter, ρ_e , and the dose per fraction, d_a or d_b , be required to attain the CRE appropriate to a particular stage of the regime, are given by $n_{a_1}, n_{a_2}, n_{a_3}, \dots, n_{a_N}$, and $n_{b_1}, n_{b_2}, n_{b_3}, \dots, n_{b_N}$, respectively. In general, the values of $n_{a_1}, \dots, n_{a_N}$, and $n_{b_1}, \dots, n_{b_N}$ will not be integral. The mean-

ing of this symbolism can perhaps be more clearly seen by reference to Figure 5.

If the first fraction given is of schedule A, the CRE achieved will be that due to the fraction denoted by n_{a_1} , where n_{a_1} has, in this particular instance, the value 1. The next fraction, the first of schedule B, will be that from $(n_{b_1} - 1)$ to n_{b_1} , the difference being unity, where n_{a_1} and $(n_{b_1} - 1)$ of schedules A and B, respectively, would each achieve the same CRE. This argument can be continued for succeeding fractions. Each fraction must add to the accumulated effect of previous fractions and the radiation effects at the junctions of these fractions can be related by a series of equivalence conditions:

$$\begin{aligned} \rho_e \cdot d_a \cdot n_{a_1}^m &= \rho_e \cdot d_b \cdot (n_{b_1} - 1)^m \\ \rho_e \cdot d_a \cdot (n_{a_2} - 1)^m &= \rho_e \cdot d_b \cdot n_{b_1}^m \\ \rho_e \cdot d_a \cdot n_{a_2}^m &= \rho_e \cdot d_b \cdot (n_{b_2} - 1)^m \\ &\vdots \\ \rho_e \cdot d_a \cdot n_{a_N}^m &= \rho_e \cdot d_b \cdot (n_{b_N} - 1)^m. \end{aligned}$$

It was shown in Section 3 that the CRE can be represented as a summation of effects following on each other, so it can be written as:

$$\begin{aligned} R_F &= \rho_e \{d_a \cdot n_{a_1}^m + d_b \cdot [n_{b_1}^m - (n_{b_1} - 1)^m] \\ &\quad + d_a \cdot [n_{a_2}^m - (n_{a_2} - 1)^m] \dots + d_b \cdot [n_{b_N}^m - (n_{b_N} - 1)^m]\}. \end{aligned}$$

Using the equivalence relations established above, and cancelling equal terms, R_F can be written as:

$$R_F = \rho_e \cdot d_b \cdot n_{b_N}^m.$$

The problem now remains to relate n_{b_N} to the fractionations of schedules A and B. This can be done by resorting to the equivalence conditions noted above. For convenience, let $y = (d_a/d_b)^{1/m}$. The equivalence conditions can then be written in the simpler form:

$$\begin{aligned} y \cdot n_{a_1} &= n_{b_1} - 1 \\ y \cdot (n_{a_2} - 1) &= n_{b_1} \\ y \cdot n_{a_2} &= n_{b_2} - 1 \\ &\vdots \\ y \cdot n_{a_N} &= n_{b_N} - 1 \end{aligned}$$

As a particular example,

$$\begin{aligned} n_{b_2} &= y \cdot n_{a_2} + 1 = y \cdot \left[\frac{n_{b_1}}{y} + 1 \right] + 1 \\ &= y \left[\frac{y \cdot n_{a_1} + 1}{y} + 1 \right] + 1 \\ &= y \cdot n_{a_1} + 1 + y + 1 = y(n_{a_1} + 1) + 2. \end{aligned}$$

By repeated application of this method, any of the $n_{b_1}, \dots, n_{b_N}$, can be put in terms of n_{a_1} , the first fraction given, which is also identified with the first fraction of schedule A whose contribution to the CRE is readily found. By this repeated application, it can be shown that:

$$\begin{aligned} n_{b_1} &= y \cdot n_{a_1} + 1 \\ n_{b_2} &= y \cdot (n_{a_1} + 1) + 2 \\ n_{b_3} &= y \cdot (n_{a_1} + 2) + 3 \\ &\vdots \\ n_{b_N} &= y \cdot [n_{a_1} + (N-1)] + N. \end{aligned}$$

Replacing y by $(d_a/d_b)^{1/m}$, the CRE can be written as:

$$\begin{aligned} R_F &= \rho_e \cdot d_b \cdot \{(d_a/d_b)^{1/m} \cdot [n_{a_1} + (N-1)] + N\}^m \\ &= \rho_e \cdot \{d_a^{1/m} \cdot [n_{a_1} + (N-1)] + d_b^{1/m} \cdot N\}^m. \end{aligned}$$

Now, since N is the number of treatments given on any schedule, then $N = N_a = N_b$. If $\mathcal{N}$ is the total number of treatments, then $\mathcal{N} = 2N$. Also, as noted earlier, $n_{a_1} = 1$, so that

$$\begin{aligned} R_F &= \rho_e \cdot N^m \cdot (d_a^{1/m} + d_b^{1/m})^m \\ &= \rho_e \cdot \mathcal{N}^m \cdot \left[\frac{d_a^{1/m} + d_b^{1/m}}{2} \right]^m. \end{aligned}$$

From the symmetry of this result, it is clear that the same CRE is achieved regardless of whether the first fraction is from schedule A or B.

Appendix 3: General Proof for the Equivalence of Permuted Fractions

Consider the situation in which a certain CRE has been achieved by some arbitrary sub-tolerance treatment regime. It is required to compare the increases in the CRE from that value by the application of 2 different fractions first in one order and then in the reverse order. Suppose that the interval between fractions is constant so that a constant repetition parameter, ρ , can be assumed. Consider, first, that the 2 additional fractions are to be given in the order d_a followed by d_b . Suppose that the initial value of the CRE could be achieved by $(n_a - 1)$ fractions of the schedule A on which the first additional fraction is to be given. At the junction between the added fractions, the following equivalence condition must hold:

$$\rho \cdot d_a \cdot n_a^m = \rho \cdot d_b \cdot (n_b - 1)^m,$$

where n_b is the number of fractions on schedule B, to which the second additional fraction would belong, to attain the ultimate CRE.

Let ΔR_1 be the increment in CRE from the two additional fractions. Then,

$$\begin{aligned} \Delta R_1 &= \rho \cdot \{[d_a \cdot n_a^m - d_a \cdot (n_a - 1)^m] \\ &\quad + [d_b \cdot n_b^m - d_b \cdot (n_b - 1)^m]\}. \end{aligned}$$

From the equivalence condition at the junction point,

$$\Delta R_1 = \rho \cdot \{d_b \cdot n_b^m - d_a \cdot (n_a - 1)^m\}.$$

This increment in the CRE, ΔR_1 , can be put in a more convenient form for comparison with the expression for the application of the fractions in reverse order by a further application of the equivalence condition giving:

$$\begin{aligned} (d_a/d_b)^{1/m} &= \frac{n_b - 1}{n_a}, \text{ or } n_b = 1 + n_a \cdot (d_a/d_b)^{1/m}. \\ \therefore \Delta R_1 &= \rho \cdot \{d_b \cdot [1 + n_a \cdot (d_a/d_b)^{1/m}]^m \\ &\quad - d_a \cdot (n_a - 1)^m\} \\ &= \rho \cdot \{[d_b^{1/m} + n_a \cdot d_a^{1/m}]^m - d_a \cdot (n_a - 1)^m\}. \end{aligned}$$

Now, consider the application of those same 2 fractions in the reverse order. Suppose that the initial value of the CRE could be achieved by $(n'_b - 1)$ fractions of schedule B, on which the first additional fraction is now to be given. At the junction between the added fractions, the equivalence condition is now:

$$\rho \cdot d_b \cdot n'_b{}^m = \rho \cdot d_a \cdot (n'_a - 1)^m,$$

where n'_a is the number of fractions on schedule A, to which the second additional fraction now belongs, to give the ultimate CRE. Let ΔR_2 be the increment in CRE from the 2 additional fractions in the present order. Then,

$$\begin{aligned} \Delta R_2 &= \rho \cdot \{[d_b \cdot n'_b{}^m - d_b \cdot (n'_b - 1)^m] \\ &\quad + [d_a \cdot n'_a{}^m - d_a \cdot (n'_a - 1)^m]\}. \end{aligned}$$

By again applying the equivalence condition,

$$\Delta R_2 = \rho \cdot \{d_a \cdot n'_a{}^m - d_b \cdot (n'_b - 1)^m\},$$

$$\text{and } \Delta R_2 = \rho \cdot \{[d_a^{1/m} + n'_b \cdot d_b^{1/m}]^m - d_b \cdot (n'_b - 1)^m\}.$$

At the junction of the original regime with the first additional fraction, a further equivalence condition can be established for both orders of application and s:

$$\begin{aligned} \rho \cdot d_a \cdot (n_a - 1)^m &= \rho \cdot d_b \cdot (n'_b - 1)^m \\ \therefore \frac{n'_b - 1}{n_a - 1} &= (d_a/d_b)^{1/m} \\ \therefore n'_b &= 1 + (n_a - 1) \cdot (d_a/d_b)^{1/m}. \end{aligned}$$

Thus,

$$\begin{aligned}\Delta R_2 &= \rho \cdot \{ [d_a^{1/m} + \{1 + (n_a - 1) \cdot (d_a/d_b)^{1/m}\} \cdot d_b^{1/m}]^m \\ &\quad - d_a \cdot (n_a - 1)^m \} \\ &= \rho \cdot \{ [d_a^{1/m} + d_b^{1/m} + (n_a - 1) \cdot d_a^{1/m}]^m \\ &\quad - d_a \cdot (n_a - 1)^m \} \\ &= \rho \cdot \{ [d_b^{1/m} + n_a \cdot d_a^{1/m}]^m - d_a \cdot (n_a - 1)^m \} \\ &= \Delta R_1.\end{aligned}$$

This equality demonstrates that the order of treatments given at some arbitrary point in a course of radiation therapy is of no consequence to the accumulated radiation damage achieved.

Although this result follows more directly from Ellis' approach (since he defines each fraction of a given schedule to have the same effect), this proof has been included for the sake of completeness of discussion of the CRE system, which, in general,

permits a more logical step-by-step method of evaluating the effect of complex regimes.

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BOOK REVIEW

Atlas der Szintigraphie. By H. OESER, W. SCHUMACHER, H. ERNST and D. FROST. Published by W. de Gruyter & Co., Berlin, 1969. DM. 190.

This book deals thoroughly with its subject. One attractive feature is its unusual format. All the leaves are loose and mounted together within plastic-covered hard backs. There is space to add many more leaves within this framework. The text (in German) is divided into fifteen sections, labelled so that each can be found readily and may be removed easily. These sections discuss in turn basic nuclear physics, image recording, dosimetry and the applications of scintigraphy to each organ.

The earlier, technical sections are illustrated profusely with good quality line diagrams as well as photographs of individual scanners. They consider many problems in detail. Image display and subtraction are discussed and well illustrated.

The first clinical section deals with the thyroid gland. After a brief historical table relevant to the thyroid gland, the radionuclides in general use, their physical characteristics and dosimetry are discussed. All the available methods of displaying the image of a thyroid phantom—in black and white, in colour, by camera scintiphoto, and by isointensity trace—are illustrated beautifully on high quality paper. The clinical applications of thyroid scintigraphy are discussed systematically with several illustrative cases, and abundant radiograms and scintigrams. The section ends with a full list of brief references from the world literature. The 64 pages devoted to the thyroid gland are all that one would expect from an Atlas of Scintigraphy.

The remaining eleven sections are written to the same pattern and together cover all the standard clinical applications of radionuclides. It is understandable that the long-standing interest and copious knowledge of the thyroid gland have gained it pride of place in this book, but it is a pity that the authors have not dealt with some other sections in greater detail, particularly in a book of this quality and price. For example the section on the lungs is as short as that on the cardio-vascular system, and the sections on the whole of the nervous system are not much longer than that on the lymphatic system. This apparent imbalance may only reflect the emphasis on different interests in Germany and in this country; but it was disappointing not to find more of the excellent illustrations of pulmonary and cerebral lesions, not because conciseness is any disadvantage, but because an "Atlas" is expected to be something of a library of illustrations.

The renal section is a good example of a well-balanced account with profuse scintiphotos showing how well the gamma camera records of dynamic studies. Linguistic purists will take pleasure from seeing "renogram" supplanted by "nephrogram" and perhaps regret the precedence enjoyed, in English, by the "urographic nephrogram".

This atlas was first published in 1969 and inevitably the most recent references it contains are from 1967-8. This does not diminish its value as an atlas of standard and accepted scintigraphy but the rapid annual advances in this field may help to persuade the authors of this first-class book to update it regularly. If the next edition or supplement appeared in English also, it would reach a large number of receptive readers.

E. RHYS DAVIES